Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: John Brasil Dairy #3

Physical address of dairy:

Number and Street	City	County	Zip Code
1707 S Mitchell RD	Turlock	Merced	95380

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 11/01/1991

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

X058-X011-X011-XXXX X058-X015-X008-XXXX X058-X016-X016-XXXX X058-X030-X007-XXXX

B. OPERATORS

Brasil, John

Operator name: Brasil, John

Telephone no.: (209) 632-7867 (Landline) (Cellular)

Mailing Address Number and Street	City	State	Zip Code
2613 S Mitchell RD	Turlock	CA	95380

This operator is responsible for paying permit fees.

C. OWNERS

Brasil, John

Legal owner name: Brasil, John

Telephone no.: (209) 632-7867 (Landline) (Cellular)

Mailing Address Number and Street	City	State	Zip Code
2613 S Mitchell RD	Turlock	CA	95380

This owner is responsible for paying permit fees.

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	63	0	0	0	0
Number under roof	578	0	0	0	0	0
Maximum number	578	63	0	0	0	0
Average number	578	63	0	0	0	0
Avg live weight (lbs)	1,200	1,300	0	0		

Predominant milk cow breed: Jersey

Average milk production: 56 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 14,753.61 *tons per reporting period*

Total nitrogen from manure: 192,930.35 *lbs per reporting period*

After ammonia losses (30% loss applied): 135,051.25 *lbs per reporting period*

Total phosphorus from manure: 32,052.14 *lbs per reporting period*

Total potassium from manure: 105,375.89 *lbs per reporting period*

Total salt from manure: 286,638.15 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 13,077,000 *gallons*

Total nitrogen generated: 74,721.66 *lbs*

Total phosphorus generated: 14,187.48 *lbs*

Total potassium generated: 123,459.85 *lbs*

Total salt generated: 562,089.58 *lbs*

	13,077,000	*gallons applied*
+	0	*gallons exported*
-	0	*gallons imported*
=	13,077,000	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
Domestic Well - Barn Well	Ground water
Domestic Well - House Well	Ground water
Irrigation Water - Canal	Surface water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

No process wastewater nutrient imports entered.

No commercial or other nutrient imports entered.

G. NUTRIENT EXPORTS

Date	Material type	Quantity	Reporting basis	Moisture (%)	Density (lbs/cu ft)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/19/2023	Corral solids	850.00 *ton*	Dry-weight	8.0		32,700.00	7,800.00	22,500.00		0.00

No liquid nutrient exports entered.

Material type	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Dry manure	51,142.80	12,199.20	35,190.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Total exports for all materials	51,142.80	12,199.20	35,190.00	0.00

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
Dairy Field	63	63	2	process wastewater	X058-X016-X016-XXXX
Lagoon Field	16	16	2	process wastewater	X058-X015-X008-XXXX
Mulders	58	58	2	process wastewater	X058-X015-X012-XXXX
Totals for areas that were used for application	137	137	6		
Totals for areas that were not used for application					
Land application area totals	137	137	6		

B. CROPS AND HARVESTS

Dairy Field

Field name: Dairy Field

11/06/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 63 Plant date: 11/06/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/25/2023	597.43 *ton*	Dry-weight		63.8	25,300.00	5,100.00	42,800.00		11.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	9.48	173.70	35.02	293.85	796.42

06/06/2023: Corn, silage

Crop: Corn, silage Acres planted: 63 Plant date: 06/06/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/20/2023	1,944.71 *ton*	Dry-weight		70.7	12,900.00	2,400.00	12,000.00		4.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	30.87	233.35	43.41	217.07	814.00

Lagoon Field

Field name: Lagoon Field

11/06/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 16 Plant date: 11/06/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/25/2023	144.25 *ton*	Dry-weight		62.2	26,100.00	5,000.00	40,600.00		12.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	9.02	177.89	34.08	276.72	845.16

06/06/2023: Corn, silage

Crop: Corn, silage Acres planted: 16 Plant date: 06/06/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/20/2023	487.55 *ton*	Dry-weight		68.3	12,600.00	2,600.00	13,100.00		4.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	30.47	243.42	50.23	253.08	946.64

Mulders

Field name: Mulders

11/06/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 58 Plant date: 11/06/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/25/2023	550.41 *ton*	Dry-weight		64.5	26,400.00	5,000.00	42,300.00		11.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	9.49	177.88	33.69	285.01	781.58

Mulders

06/06/2023: Corn, silage

Crop: Corn, silage Acres planted: 58 Plant date: 06/06/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/20/2023	1,802.58 *ton*	Dry-weight		71.1	13,900.00	2,800.00	13,400.00		5.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	31.08	249.69	50.30	240.71	916.15

NUTRIENT BUDGET

A. LAND APPLICATIONS

Dairy Field - 11/06/2022: Oats, silage-soft dough

Field name: Dairy Field

Crop: Oats, silage-soft dough — Plant date: 11/06/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/31/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	109.88	18.08	182.34	810.22	1,197,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	51.92	9,799,425.00 *gal*
Application event totals		109.88	18.08	182.34	862.14	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/22/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	107.40	17.67	178.23	791.94	1,170,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	50.49	9,530,025.00 *gal*
Application event totals		107.40	17.67	178.23	842.43	

Dairy Field - 06/06/2023: Corn, silage

Field name: Dairy Field

Crop: Corn, silage — Plant date: 06/06/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	49.78	9,395,325.00 *gal*
Application event totals		0.00	0.00	0.00	49.78	

Dairy Field - 06/06/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	49.42	9,327,975.00 *gal*
Application event totals		0.00	0.00	0.00	49.42	

07/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	49.78	9,395,325.00 *gal*
Application event totals		0.00	0.00	0.00	49.78	

07/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	107.40	17.67	178.23	791.94	1,170,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	50.32	9,496,350.00 *gal*
Application event totals		107.40	17.67	178.23	842.26	

08/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	108.23	17.80	179.60	798.03	1,179,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	50.14	9,462,675.00 *gal*
Application event totals		108.23	17.80	179.60	848.17	

08/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	103.27	16.99	171.37	761.48	1,125,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	49.78	9,395,325.00 *gal*
Application event totals		103.27	16.99	171.37	811.26	

Dairy Field - 06/06/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	49.25	9,294,300.00 *gal*
Application event totals		0.00	0.00	0.00	49.25	

Lagoon Field - 11/06/2022: Oats, silage-soft dough

Field name: Lagoon Field

Crop: Oats, silage-soft dough Plant date: 11/06/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/31/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	113.85	42.75	182.79	650.60	297,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	54.10	2,592,975.00 *gal*
Application event totals		113.85	42.75	182.79	704.69	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	103.50	38.87	166.17	591.45	270,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	44.26	2,121,525.00 *gal*
Application event totals		103.50	38.87	166.17	635.71	

Lagoon Field - 06/06/2023: Corn, silage

Field name: Lagoon Field

Crop: Corn, silage Plant date: 06/06/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

Lagoon Field - 06/06/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	55.50	2,660,325.00 *gal*
Application event totals		0.00	0.00	0.00	55.50	

07/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	54.10	2,592,975.00 *gal*
Application event totals		0.00	0.00	0.00	54.10	

07/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	52.69	2,525,625.00 *gal*
Application event totals		0.00	0.00	0.00	52.69	

07/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	148.36	55.71	238.18	847.75	387,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	55.50	2,660,325.00 *gal*
Application event totals		148.36	55.71	238.18	903.25	

08/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	100.05	37.57	160.63	571.74	261,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	54.10	2,592,975.00 *gal*
Application event totals		100.05	37.57	160.63	625.83	

Lagoon Field - 06/06/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	86.25	32.39	138.47	492.88	225,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	52.69	2,525,625.00 *gal*
Application event totals		86.25	32.39	138.47	545.57	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	55.50	2,660,325.00 *gal*
Application event totals		0.00	0.00	0.00	55.50	

Mulders - 11/06/2022: Oats, silage-soft dough

Field name: Mulders

Crop: Oats, silage-soft dough

Plant date: 11/06/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/31/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	113.97	18.75	189.12	840.36	1,143,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	47.48	8,250,375.00 *gal*
Application event totals		113.97	18.75	189.12	887.84	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	102.30	16.83	169.76	754.34	1,026,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	47.09	8,183,025.00 *gal*
Application event totals		102.30	16.83	169.76	801.43	

Mulders - 06/06/2023: Corn, silage

Mulders - 06/06/2023: Corn, silage

Field name: Mulders

Crop: Corn, silage

Plant date: 06/06/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	47.48	8,250,375.00 *gal*
Application event totals		0.00	0.00	0.00	47.48	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	47.87	8,317,725.00 *gal*
Application event totals		0.00	0.00	0.00	47.87	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	47.48	8,250,375.00 *gal*
Application event totals		0.00	0.00	0.00	47.48	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	112.05	18.72	185.39	1,058.36	1,251,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	47.09	8,183,025.00 *gal*
Application event totals		112.05	18.72	185.39	1,105.45	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	122.94	20.22	204.01	906.53	1,233,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	47.87	8,317,725.00 *gal*
Application event totals		122.94	20.22	204.01	954.40	

Mulders - 06/06/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	102.37	17.10	169.39	966.99	1,143,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	47.48	8,250,375.00 *gal*
Application event totals		102.37	17.10	169.39	1,014.47	

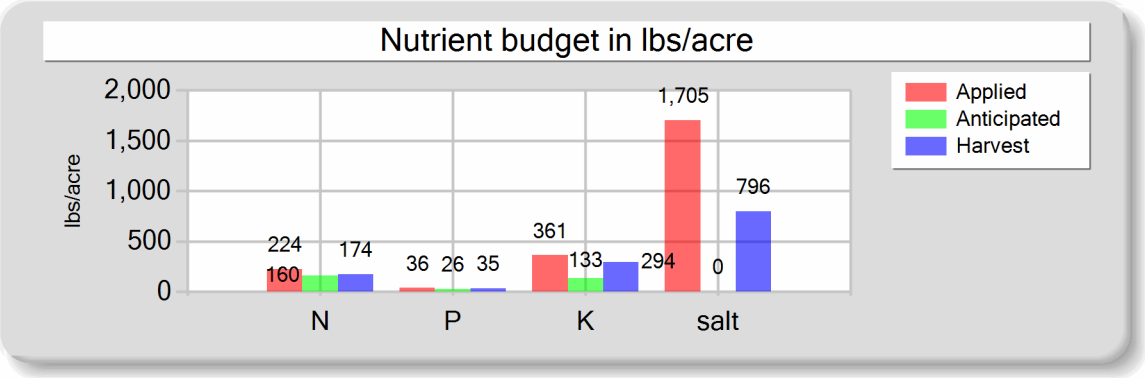
09/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	47.09	8,183,025.00 *gal*
Application event totals		0.00	0.00	0.00	47.09	

B. NUTRIENT BUDGET

Dairy Field - 11/06/2022: Oats, silage-soft dough

Field name: Dairy Field Crop: Oats, silage-soft dough Plant date: 11/06/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	217.28	35.74	360.56	1,602.16
Fresh water	0.00	0.00	0.00	102.42
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	224.28	35.74	360.56	1,704.57
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	173.70	35.02	293.85	796.42
Nutrient balance	50.58	0.73	66.71	908.15
Applied to removed ratio	1.29	1.02	1.23	2.14

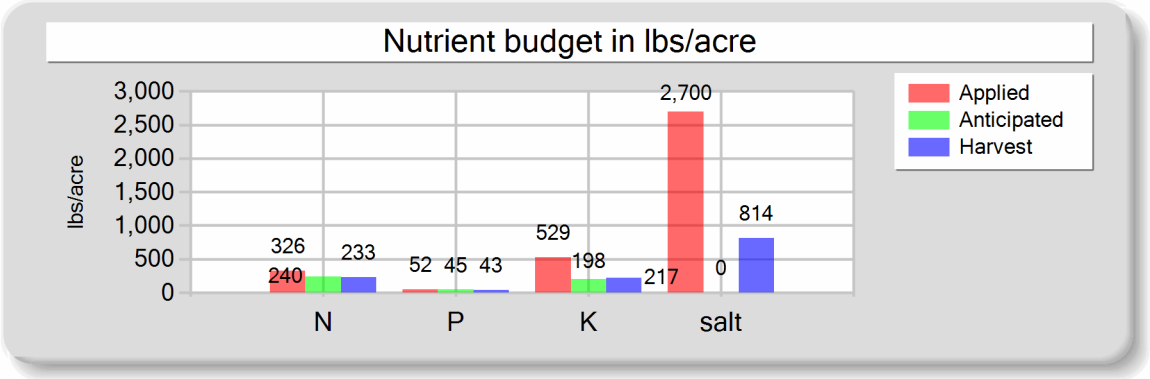
Fresh water applied	
19,329,450.00	*gallons*
711.84	*acre-inches*
11.30	*inches/acre*

Process wastewater applied	
2,367,000.00	*gallons*
87.17	*acre-inches*
1.38	*inches/acre*

Total harvests for the crop	
1	*harvests*

Dairy Field - 06/06/2023: Corn, silage

Field name: Dairy Field Crop: Corn, silage Plant date: 06/06/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	318.90	52.46	529.19	2,351.45
Fresh water	0.00	0.00	0.00	348.46
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	325.90	52.46	529.19	2,699.92
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	233.35	43.41	217.07	814.00
Nutrient balance	92.55	9.05	312.13	1,885.92
Applied to removed ratio	1.40	1.21	2.44	3.32

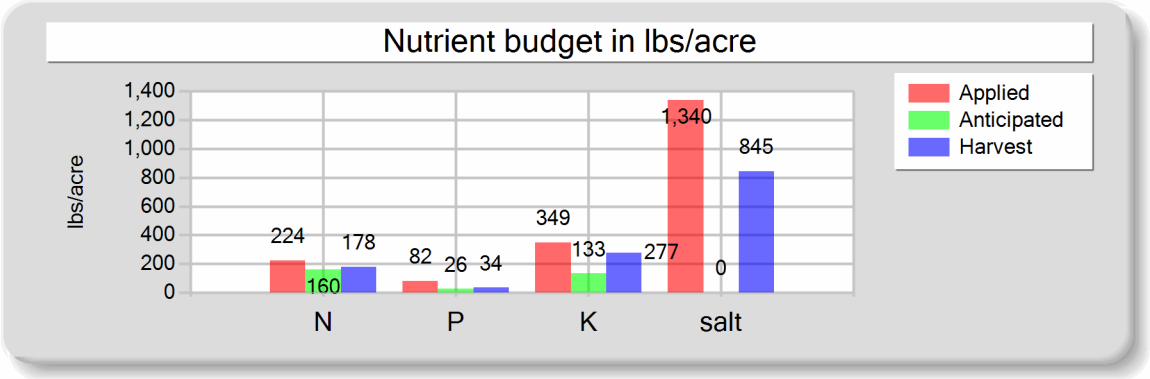
Fresh water applied	
65,767,275.00	*gallons*
2,421.99	*acre-inches*
38.44	*inches/acre*

Process wastewater applied	
3,474,000.00	*gallons*
127.94	*acre-inches*
2.03	*inches/acre*

Total harvests for the crop	
1	*harvests*

Lagoon Field - 11/06/2022: Oats, silage-soft dough

Field name: Lagoon Field Crop: Oats, silage-soft dough Plant date: 11/06/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	217.36	81.62	348.96	1,242.05
Fresh water	0.00	0.00	0.00	98.36
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	224.36	81.62	348.96	1,340.41
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	177.89	34.08	276.72	845.16
Nutrient balance	46.47	47.54	72.23	495.24
Applied to removed ratio	1.26	2.40	1.26	1.59

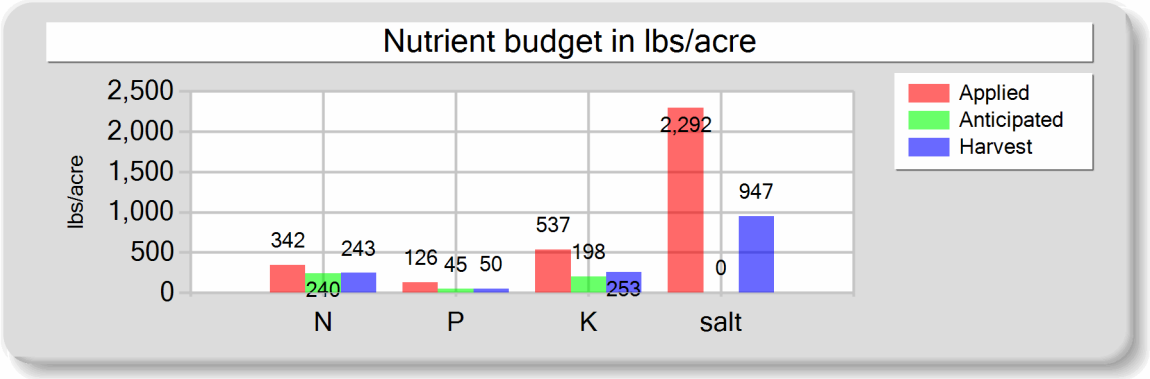
Fresh water applied	
4,714,500.00	*gallons*
173.62	*acre-inches*
10.85	*inches/acre*

Process wastewater applied	
567,000.00	*gallons*
20.88	*acre-inches*
1.31	*inches/acre*

Total harvests for the crop	
1	*harvests*

Lagoon Field - 06/06/2023: Corn, silage

Field name: Lagoon Field | Crop: Corn, silage | Plant date: 06/06/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	334.66	125.67	537.28	1,912.36
Fresh water	0.00	0.00	0.00	380.08
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	341.66	125.67	537.28	2,292.44
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	243.42	50.23	253.08	946.64
Nutrient balance	98.24	75.44	284.20	1,345.80
Applied to removed ratio	1.40	2.50	2.12	2.42

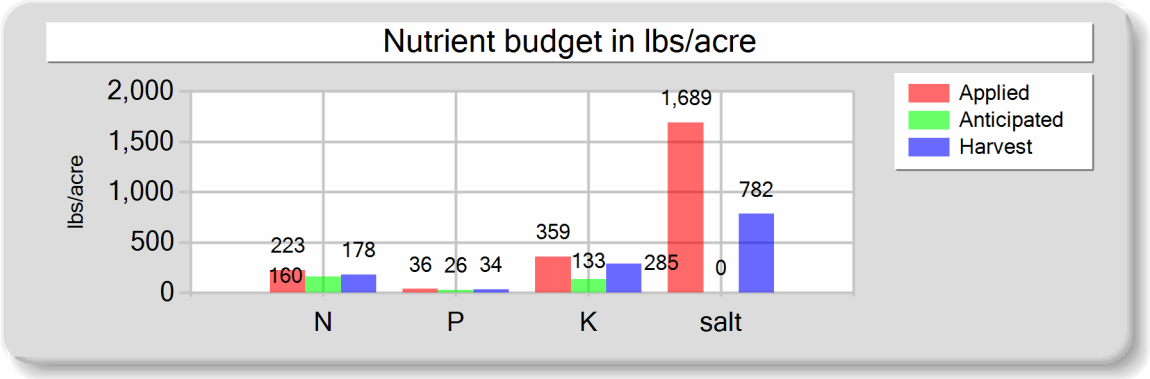
Fresh water applied	
18,218,175.00	*gallons*
670.91	*acre-inches*
41.93	*inches/acre*

Process wastewater applied	
873,000.00	*gallons*
32.15	*acre-inches*
2.01	*inches/acre*

Total harvests for the crop	
1	*harvests*

Mulders - 11/06/2022: Oats, silage-soft dough

Field name: Mulders Crop: Oats, silage-soft dough Plant date: 11/06/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	216.27	35.58	358.89	1,594.70
Fresh water	0.00	0.00	0.00	94.58
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	223.27	35.58	358.89	1,689.28
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	177.88	33.69	285.01	781.58
Nutrient balance	45.39	1.89	73.88	907.69
Applied to removed ratio	1.26	1.06	1.26	2.16

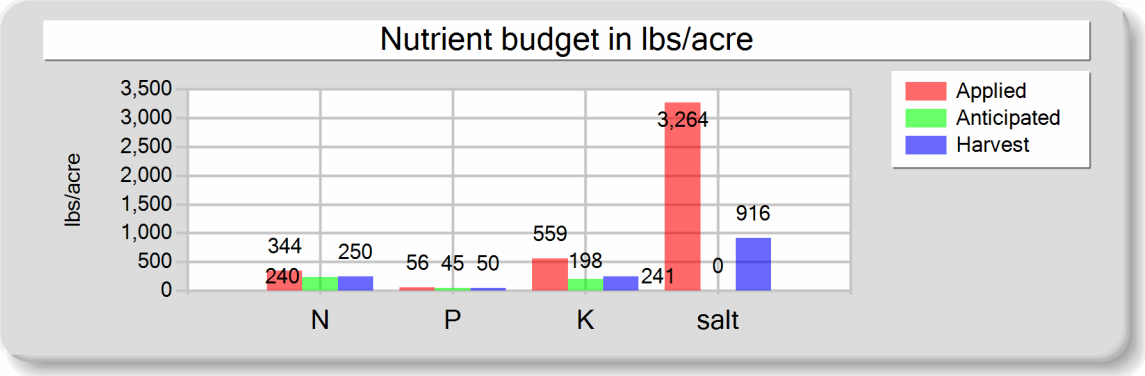
Fresh water applied	
16,433,400.00	*gallons*
605.19	*acre-inches*
10.43	*inches/acre*

Process wastewater applied	
2,169,000.00	*gallons*
79.88	*acre-inches*
1.38	*inches/acre*

Total harvests for the crop	
1	*harvests*

Mulders - 06/06/2023: Corn, silage

Field name: Mulders | Crop: Corn, silage | Plant date: 06/06/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	337.36	56.05	558.79	2,931.88
Fresh water	0.00	0.00	0.00	332.38
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	344.36	56.05	558.79	3,264.26
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	249.69	50.30	240.71	916.15
Nutrient balance	94.66	5.75	318.08	2,348.11
Applied to removed ratio	1.38	1.11	2.32	3.56

Fresh water applied	
57,752,625.00	*gallons*
2,126.83	*acre-inches*
36.67	*inches/acre*

Process wastewater applied	
3,627,000.00	*gallons*
133.57	*acre-inches*
2.30	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

Solid Manure

Sample and source description: Solid Manure

Sample date: 06/05/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 8.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	32,700.00	7,800.00	22,500.00							
DL	100.00	100.00	100.00							

Solid Manure

Sample and source description: Solid Manure

Sample date: 10/16/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 8.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	15,700.00	4,900.00	24,200.00							
DL	100.00	100.00	100.00							

B. PROCESS WASTEWATER ANALYSES

Lagoon Water

Sample and source description: Lagoon Water

Sample date: 02/20/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.20

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	733.00	471.00	0.00	2.00	276.00	1,180.00								11,500.00	4,200
DL	1.00	0.50	0.50	0.10	0.10	0.50								10.00	10

Waste water

Sample and source description: Waste water

Sample date: 06/14/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.50

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	692.00	353.00	0.00	1.00	114.00	1,150.00								10,700.00	5,110
DL	1.00	0.50	0.50	0.10	0.10	0.50								10.00	10

Lagoon Water

Sample and source description: Lagoon Water

Sample date: 09/25/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.60

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	622.00	453.00	0.00	0.50	104.00	1,030.00								9,100.00	5,880
DL	1.00	0.50	0.50	0.10	0.10	0.50								10.00	10

Lagoon Water

Sample and source description: Lagoon Water

Sample date: 11/13/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.30

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	1,000.00	222.00	0.00	0.00	167.00	1,320.00								8,690.00	2,830
DL	1.00	0.50	0.50	0.50	0.10	0.50								10.00	10

C. FRESH WATER ANALYSES

Domestic Well - Barn Well

Domestic Well - Barn Well

Barn Well

Sample description: Barn Well

Sample date: 11/13/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value			63.20								1,810.00	
DL			0.10								10.00	

Domestic Well - House Well

House Well

Sample description: House Well

Sample date: 11/13/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value			63.70								1,830.00	
DL			0.10								10.00	

Irrigation Water - Canal

Canal

Sample description: Canal

Sample date: 07/17/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.00										32.80	40
DL	1.00										10.00	10

D. SOIL ANALYSES

Dairy Field

Dairy Field

Sample and source description: Dairy Field

Sample date: 11/29/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			82.00				
DL			2.00				

Lagoon Field

Lagoon Field

Sample and source description: Lagoon Field

Sample date: 11/29/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			106.00				
DL			2.00				

Mulders

Mulders

Sample and source description: Mulders

Sample date: 11/29/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			129.00				
DL			2.00				

E. PLANT TISSUE ANALYSES

Dairy Field - 11/06/2022: Oats, silage-soft dough

Dairy Field - 11/06/2022: Oats, silage-soft dough

Dairy Field A

Sample and source description: Dairy Field A

Sample date: 04/25/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 63.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	25,300.00	5,100.00	42,800.00		11.60
DL	100.00	100.00	100.00		0.01

Dairy Field B

Sample and source description: Dairy Field B

Sample date: 04/25/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 60.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	24,000.00	4,600.00	38,500.00		11.90
DL	100.00	100.00	100.00		0.01

Dairy Field - 06/06/2023: Corn, silage

Dairy Field A

Sample and source description: Dairy Field A

Sample date: 09/20/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 71.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,600.00	2,500.00	11,900.00		4.60
DL	100.00	100.00	100.00		0.01

Dairy Field - 06/06/2023: Corn, silage

Dairy Field B

Sample and source description: Dairy Field B

Sample date: 09/20/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 70.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,900.00	2,400.00	12,000.00		4.50
DL	100.00	100.00	100.00		0.01

Lagoon Field - 11/06/2022: Oats, silage-soft dough

Lagoon Field

Sample and source description: Lagoon Field

Sample date: 04/25/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 62.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	26,100.00	5,000.00	40,600.00		12.40
DL	100.00	100.00	100.00		0.01

Lagoon Field - 06/06/2023: Corn, silage

Lagoon Field

Sample and source description: Lagoon Field

Sample date: 09/20/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 68.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,600.00	2,600.00	13,100.00		4.90
DL	100.00	100.00	100.00		0.01

Mulders - 11/06/2022: Oats, silage-soft dough

Mulders Field A

Sample and source description: Mulders Field A

Sample date: 04/25/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 64.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	26,400.00	5,000.00	42,300.00		11.60
DL	100.00	100.00	100.00		0.01

Mulders Field B

Sample and source description: Mulders Field B

Sample date: 04/25/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 64.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	27,600.00	5,100.00	42,100.00		11.50
DL	100.00	100.00	100.00		0.01

Mulders - 06/06/2023: Corn, silage

Mulders Field A

Sample and source description: Mulders Field A

Sample date: 09/20/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 71.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,900.00	2,800.00	13,400.00		5.10
DL	100.00	100.00	100.00		0.01

Mulders - 06/06/2023: Corn, silage

Mulders Field B

Sample and source description: Mulders Field B

Sample date: 09/20/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 69.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	11,200.00	2,500.00	12,400.00		4.50
DL	100.00	100.00	100.00		0.01

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

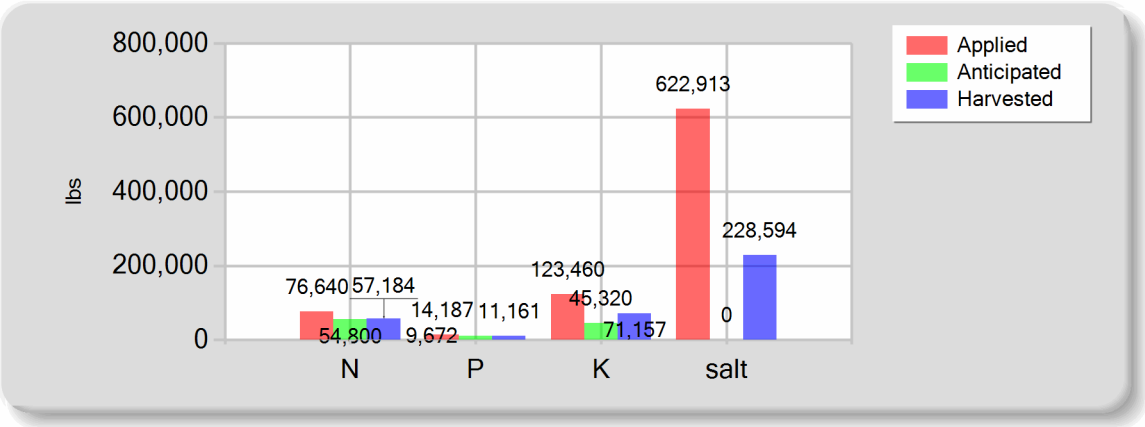
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

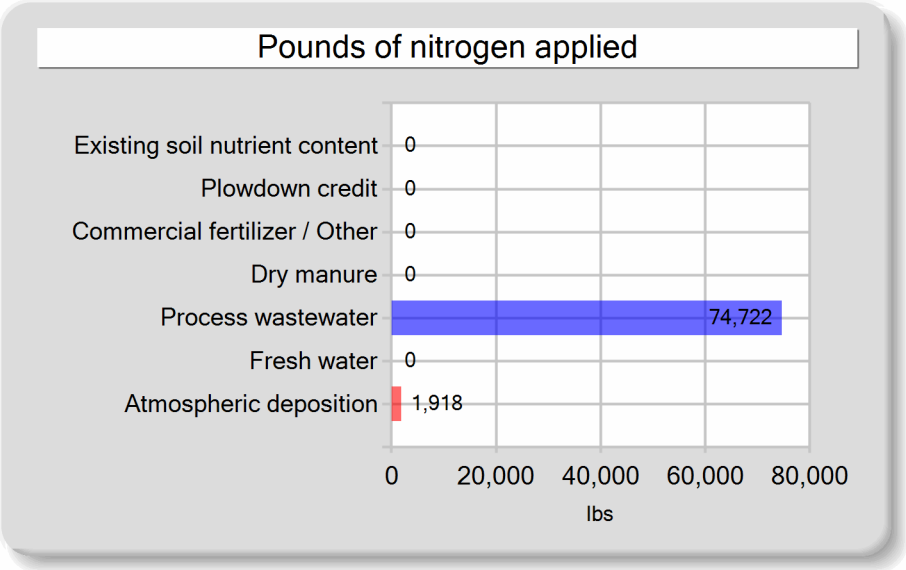
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

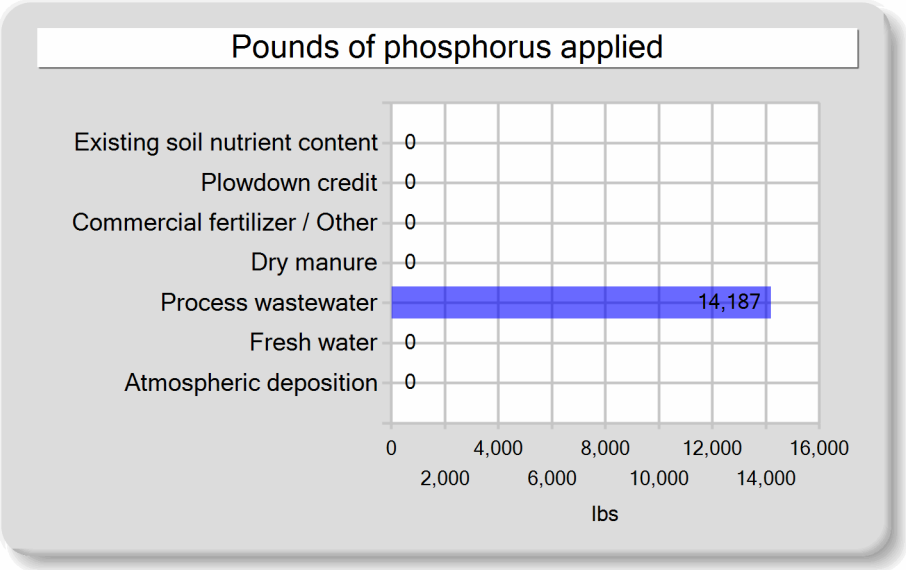
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	74,721.66	14,187.48	123,459.85	562,089.58
Fresh water	0.00	0.00	0.00	60,823.51
Atmospheric deposition	1,918.00	0.00	0.00	0.00
Total nutrients applied	76,639.66	14,187.48	123,459.85	622,913.08
Anticipated crop nutrient removal	54,800.00	9,672.20	45,319.60	0.00
Actual crop nutrient removal	57,184.29	11,161.18	71,156.54	228,593.58
Nutrient balance	19,455.38	3,026.30	52,303.31	394,319.50
Applied to removed ratio	1.34	1.27	1.74	2.72

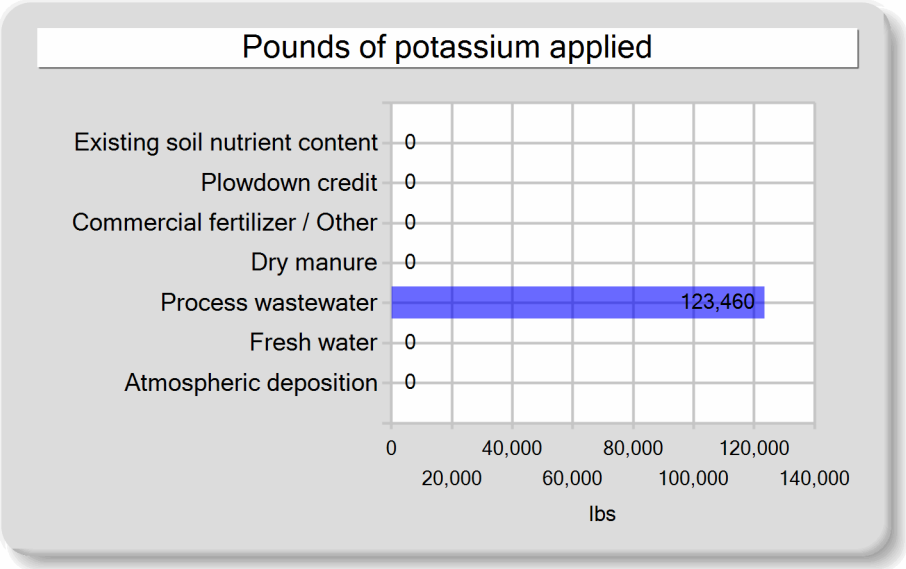
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

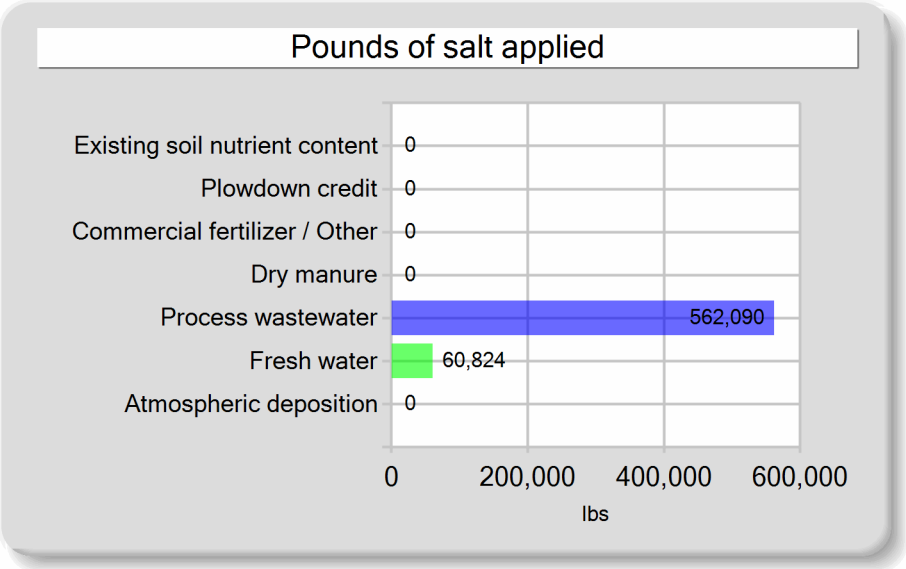


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	Yes
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

No notes entered for this annual report.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
John Brasil	SAME AS OWNER
PRINT OR TYPE NAME	PRINT OR TYPE NAME
6/28/2024	
DATE	DATE

ATTACHMENT D

Manure Tracking Manifest
For
Existing Milk Cow Dairies

Instructions:

1) Complete one manifest for each hauling event, for each destination. A hauling event may last for several days, as long as the manure is being hauled to the same destination.
2) If there are multiple destinations, **complete a separate form for each destination**.
3) The operator must obtain the signature of the hauler upon completion of each manure-hauling event.
4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

Operator Information:

Name of Operator: **John Brasil**

Name of Dairy Facility: **John Brasil Dairy #3**

Facility Address: **1707 S. Mitchell Road, Turlock, CA 95380**

Contact Person Name and Phone Number: **John Brasil (209) 652-3681**

Manure Hauler Information:

Name of Hauling Company/Person: **John Brasil Dairy #1**

Address of Hauling Company /Person: **2613 S. Mitchell Road, Turlock, CA 95380**

Contact Person and Phone Number: **John Brasil (209) 652-3681**

Destination Information:

Composting Facility / Broker / Farmer / Other (identify) **Farmer** (please circle one)

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

Name	Number and Street	City	Zip Code	Phone Number
JB Farms	8400 W. Linwood Avenue	Turlock	95380	(209) 656-9005

Manure/Process Wastewater Destination Address or Assessor's Parcel Number:

Number and Street	City	Zip Code	Assessor's Parcel Number
9730 W. Bradbury Road	Turlock	95380	

Dates Hauled: **5/19/2023**

Amount Hauled:

Enter the amount of manure hauled in tons or cubic yards (indicate the units used), the manure solids content (if amount reported in tons) or manure density (if amount reported in cubic yards), and the method used to calculate the amount:

Manure: **850** Tons

Manure Solids Content (if amount reported in tons): **92.0%**

Method used to determine amount of manure: _____

Number of Truckloads x Average Truck Weight

Enter the amount of process wastewater hauled in gallons and the method used to determine the amount.

Process Wastewater: **N/A** Gallons

Method used to determine volume of process wastewater: **N/A**

Written Agreement:

****NOT NECESSARY FOR SOLID MANURE MANIFEST****

Certification:
I declare under the penalty of law that I personally examined and am familiar with the information submitted in this document, and that based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of a fine and imprisonment for knowing violations.

Operator's Signature: _____ Date: 5/19/2023

Hauler's Signature: _____ Date: 5/19/2023

DELLAVALLE™
LABORATORY INC

R & J Avila Inc
8119 Ellington Ave
Hilmar, CA 95324

Account# 00-0019840
Account Manager: Ben Nydam
Submitted By: Rui & John Avila
Ranch: John Brasil Dairy #3

Received: 11/14/2023 15:45
Reported: 11/21/2023 09:38

Samples in this Report

Lab ID	Sample	Matrix	Sampled By	Crop	Date Sampled
23K0683-01	Barn Well	Ag Water	T. Madruga		11/13/2023 15:00
23K0683-02	House Well	Ag Water	T. Madruga		11/13/2023 16:00

Default Cooler Temperature on Receipt °C: 1.5
Containers Intact
COC/Labels Agree
Received On Ice

Notes and Definitions

Item	Definition
H	Hold Time Exceeded
MCL	Drinking Water Maximum Contaminant Level
ND	Analyte NOT DETECTED at or above the reporting limit.
NES	Not Enough Sample
*	Not Taken
RPD	Relative Percent Difference
%REC	Percent Recovery
Source	Sample that was matrix spiked or duplicated.

Scott McFriedland

Laboratory Director/Technical Manager

ELAP Certification #1595
A2LA Certification #6440.02

The results in this report apply to the samples as received and were analyzed in accordance with the chain of custody document. This analytical report must be reproduced in its entirety. All results listed in this report are for the exclusive use of the submitting party. Dellavalle Laboratory, Inc. assumes no responsibility for report alteration, separation, detachment or third party interpretation.

DELLAVALLE™
LABORATORY INC

R & J Avila Inc
8119 Ellington Ave
Hilmar, CA 95324

Account# 00-0019840
Account Manager: Ben Nydam
Submitted By: Rui & John Avila
Ranch: John Brasil Dairy #3

Received: 11/14/2023 15:45
Reported: 11/21/2023 09:38

Sample Results

Sample: Barn Well
23K0683-01 (Water)

Sampled: 11/13/2023 15:00
Sampled By: T. Madruga

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**1.81**	mmhos/cm	0.01	1		11/17/23 11:38	SM 2510 B		BEK0478
Electrical Conductivity umhos	**1810**	umhos/cm	10.0	1		11/17/23 11:38	SM 2510 B		BEK0478
Ammonia (as N)	ND	mg/L	0.00	1		11/13/23 15:00	Field		BEK0430
Nitrate Nitrogen as NO3N	**63.2**	mg/L	0.1	1	10	11/14/23 23:45	EPA 300.0		BEK0466
Temperature	**25.0**	units	0.0	1		11/17/23 11:38	SM 4500-H+	H	BEK0478
pH	**7.6**	units	1.0	1		11/17/23 11:38	SM 4500-H+	H	BEK0478

The results in this report apply to the samples as received and were analyzed in accordance with the chain of custody document. This analytical report must be reproduced in its entirety. All results listed in this report are for the exclusive use of the submitting party. Dellavalle Laboratory, Inc. assumes no responsibility for report alteration, separation, detachment or third party interpretation.

DELLAVALLE™
LABORATORY INC

R & J Avila Inc
8119 Ellington Ave
Hilmar, CA 95324

Account# 00-0019840
Account Manager: Ben Nydam
Submitted By: Rui & John Avila
Ranch: John Brasil Dairy #3

Received: 11/14/2023 15:45
Reported: 11/21/2023 09:38

Sample Results
(Continued)

Sample: House Well
23K0683-02 (Water)

Sampled: 11/13/2023 16:00
Sampled By: T. Madruga

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**1.83**	mmhos/cm	0.01	1		11/17/23 11:39	SM 2510 B		BEK0478
Electrical Conductivity umhos	**1830**	umhos/cm	10.0	1		11/17/23 11:39	SM 2510 B		BEK0478
Ammonia (as N)	ND	mg/L	0.00	1		11/13/23 16:00	Field		BEK0430
Nitrate Nitrogen as NO3N	**63.7**	mg/L	0.1	1	10	11/15/23 00:05	EPA 300.0		BEK0466
Temperature	**25.0**	units	0.0	1		11/17/23 11:39	SM 4500-H+	H	BEK0478
pH	**7.7**	units	1.0	1		11/17/23 11:39	SM 4500-H+	H	BEK0478

The results in this report apply to the samples as received and were analyzed in accordance with the chain of custody document. This analytical report must be reproduced in its entirety. All results listed in this report are for the exclusive use of the submitting party. Dellavalle Laboratory, Inc. assumes no responsibility for report alteration, separation, detachment or third party interpretation.

DELLAVALLE™ LABORATORY INC

R & J Avila Inc
8119 Ellington Ave
Hilmar, CA 95324

Account# 00-0019840
Account Manager: Ben Nydam
Submitted By: Rui & John Avila
Ranch: John Brasil Dairy #3

Received: 11/14/2023 15:45
Reported: 11/21/2023 09:38

Quality Control

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEK0466										
Blank (BEK0466-BLK1)					Prepared & Analyzed: 11/14/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEK0466-BLK2)					Prepared: 11/14/2023 Analyzed: 11/15/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEK0466-BLK3)					Prepared: 11/14/2023 Analyzed: 11/15/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
LCS (BEK0466-BS1)					Prepared: 11/14/2023 Analyzed: 11/15/2023					
Nitrate Nitrogen as NO3N	5.1		0.1	mg/L	5.000		103	90-110		
LCS (BEK0466-BS2)					Prepared: 11/14/2023 Analyzed: 11/15/2023					
Nitrate Nitrogen as NO3N	5.2		0.1	mg/L	5.000		104	90-110		
Duplicate (BEK0466-DUP1)		**Source: 23K0678-05**			Prepared: 11/14/2023 Analyzed: 11/15/2023					
Nitrate Nitrogen as NO3N	20.3		0.1	mg/L		20.3			0.133	10
Duplicate (BEK0466-DUP2)		**Source: 23K0691-01**			Prepared: 11/14/2023 Analyzed: 11/15/2023					
Nitrate Nitrogen as NO3N	31.0		0.1	mg/L		31.3			1.08	10
Matrix Spike (BEK0466-MS1)		**Source: 23K0678-05**			Prepared: 11/14/2023 Analyzed: 11/15/2023					
Nitrate Nitrogen as NO3N	25.8		0.1	mg/L	5.000	20.3	110	90-110		
Matrix Spike (BEK0466-MS2)		**Source: 23K0691-01**			Prepared: 11/14/2023 Analyzed: 11/15/2023					
Nitrate Nitrogen as NO3N	36.1		0.1	mg/L	5.000	31.3	96.8	90-110		
Reference (BEK0466-SRM1)					Prepared & Analyzed: 11/14/2023					
Nitrate Nitrogen as NO3N	10.3			mg/L	10.00		103	90-110		
Reference (BEK0466-SRM2)					Prepared: 11/14/2023 Analyzed: 11/15/2023					
Nitrate Nitrogen as NO3N	10.3			mg/L	10.00		103	90-110		
Reference (BEK0466-SRM3)					Prepared: 11/14/2023 Analyzed: 11/15/2023					
Nitrate Nitrogen as NO3N	10.4			mg/L	10.00		104	90-110		

The results in this report apply to the samples as received and were analyzed in accordance with the chain of custody document. This analytical report must be reproduced in its entirety. All results listed in this report are for the exclusive use of the submitting party. Dellavalle Laboratory, Inc. assumes no responsibility for report alteration, separation, detachment or third party interpretation.

DELLAVALLE™ LABORATORY INC

R & J Avila Inc
8119 Ellington Ave
Hilmar, CA 95324

Account# 00-0019840
Account Manager: Ben Nydam
Submitted By: Rui & John Avila
Ranch: John Brasil Dairy #3

Received: 11/14/2023 15:45
Reported: 11/21/2023 09:38

Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEK0478										
Blank (BEK0478-BLK1)					Prepared: 11/15/2023	Analyzed: 11/17/2023				
Temperature	25.0		0.0	units						
Electrical Conductivity	ND		0.01	mmhos/cm						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
pH	6.0		1.0	units						
Blank (BEK0478-BLK2)					Prepared: 11/15/2023	Analyzed: 11/17/2023				
Electrical Conductivity	ND		0.01	mmhos/cm						
Temperature	25.0		0.0	units						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
pH	7.1		1.0	units						
Blank (BEK0478-BLK3)					Prepared: 11/15/2023	Analyzed: 11/17/2023				
Electrical Conductivity	ND		0.01	mmhos/cm						
Temperature	25.0		0.0	units						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
pH	7.5		1.0	units						
Duplicate (BEK0478-DUP1)		**Source: 23K0682-02**			Prepared: 11/15/2023	Analyzed: 11/17/2023				
Electrical Conductivity	2.44		0.01	mmhos/cm		2.44			0.0205	10
pH	7.8		1.0	units		7.8			0.128	10
Electrical Conductivity umhos	2440		10.0	umhos/cm		2440			0.0205	10
Duplicate (BEK0478-DUP2)		**Source: 23K0690-02**			Prepared: 11/15/2023	Analyzed: 11/17/2023				
Electrical Conductivity	1.19		0.01	mmhos/cm		1.18			0.304	10
Electrical Conductivity umhos	1190		10.0	umhos/cm		1180			0.304	10
pH	7.8		1.0	units		7.8			0.128	10
Reference (BEK0478-SRM1)					Prepared: 11/15/2023	Analyzed: 11/17/2023				
Electrical Conductivity	562			umhos/cm	538.0		105	90-110		
Reference (BEK0478-SRM2)					Prepared: 11/15/2023	Analyzed: 11/17/2023				
pH	7.5			units	7.520		100	67021-101.32		
Reference (BEK0478-SRM3)					Prepared: 11/15/2023	Analyzed: 11/17/2023				
Electrical Conductivity	1060			umhos/cm	1000		106	90-110		
Electrical Conductivity umhos	1060			umhos/cm	1000		106	90-110		
Reference (BEK0478-SRM4)					Prepared: 11/15/2023	Analyzed: 11/17/2023				
Electrical Conductivity	1060			umhos/cm	1000		106	90-110		
Electrical Conductivity umhos	1060			umhos/cm	1000		106	90-110		
Reference (BEK0478-SRM5)					Prepared: 11/15/2023	Analyzed: 11/17/2023				
Electrical Conductivity	1080			umhos/cm	1000		108	90-110		

The results in this report apply to the samples as received and were analyzed in accordance with the chain of custody document. This analytical report must be reproduced in its entirety. All results listed in this report are for the exclusive use of the submitting party. Dellavalle Laboratory, Inc. assumes no responsibility for report alteration, separation, detachment or third party interpretation.

DELLAVALLE™
LABORATORY INC

R & J Avila Inc
8119 Ellington Ave
Hilmar, CA 95324

Account# 00-0019840
Account Manager: Ben Nydam
Submitted By: Rui & John Avila
Ranch: John Brasil Dairy #3

Received: 11/14/2023 15:45
Reported: 11/21/2023 09:38

Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEK0478 (Continued)										
Reference (BEK0478-SRM5)					Prepared: 11/15/2023	Analyzed: 11/17/2023				
Electrical Conductivity umhos	1080			umhos/cm	1000		108	90-110		
Reference (BEK0478-SRM6)					Prepared: 11/15/2023	Analyzed: 11/17/2023				
pH	4.0			units	4.000		100	97.5-102.5		
Reference (BEK0478-SRM7)					Prepared: 11/15/2023	Analyzed: 11/17/2023				
pH	4.0			units	4.000		100	97.5-102.5		
Reference (BEK0478-SRM8)					Prepared: 11/15/2023	Analyzed: 11/17/2023				
pH	4.0			units	4.000		100	97.5-102.5		

The results in this report apply to the samples as received and were analyzed in accordance with the chain of custody document. This analytical report must be reproduced in its entirety. All results listed in this report are for the exclusive use of the submitting party. Dellavalle Laboratory, Inc. assumes no responsibility for report alteration, separation, detachment or third party interpretation.

11/14/23 15:45 23K0683

DELLAVALLE LABORATORY, INC.

1910 W. McKinley Avenue, Suite 110 • Fresno, CA 93728
www.dellavallelab.com 559 233-6129 · 800 228-9896 · Fax 559 268-8174

JG

No. Samples 2 No. of Bottles: 2

Purchase Order No | Bill To: Acct # 19840 | Cons # 08

Results Need By

Water Type: [ ] Drinking Water [ ] Wastewater
[X] Ag Water [ ] Groundwater [ ] Monitoring Well

Name: R & J Avila Inc
Address: 8119 Ellington Avenue
City: Hilmar **State:** CA **Zip:** 95324
Telephone: (209) 669-7919 **Fax:** (209) 634-4604
Cell/Email: 209 678-3086 Rui / 209 678-1723 John frontdesk.randjavila@gmail.com
COPY TO:

REQUESTED BY: Rui & John Avila
PROJECT John Brasil Dairy #3
ID CROP: N/A

[X] Copy of Chain [X] QA/QC Documents
Sampled By: T. Madruga

Other:

Analysis and Bottles Required: (Please indicate Analysis)

(X) DWW1: EC, pH, NO_3-N (NH4-N Field Test)
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) DWW2: DWW1 Plus SO_4, CO_3, HCO_3, Cl, Ca, Mg, Na, TDS
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) DCW1: EC, NO_3-N, TDS + TKN & TN
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) DPW1: EC, pH, NO_3-N, NH_4-N, TKN, TDS, TP, TK
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) DPW2: DPW1 Plus SO_4, CO_3, HCO_3, Cl, Ca, Mg, Na
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) Other

	Description of Samples	Date Sampled	Time Sampled	Rec'd Temp °C	Field NH_4-N
1	Barn Well	11/13/2023	1500	1.5	ND
2	House Well	11/13/2023	1600	2.9	ND
3					
4					
5					
6					
7					
8					
9					
10					

IR Thermometer SN: 192603727
Correction Factor: 0°C
Calibration Due: 12/22/2023
Location: Laboratory

CHAIN OF CUSTODY

Carrier	Signature	Company	Received (Date/Time)	Relinquished (Date/Time)
First	John Avila	R & J Avila, Inc.	11/13/2023 @ 1600	11/14/2023 @ 900
Second	JLF	Dellavalle Laboratory, Inc.	11/14/2023 @ 900	11-14-23 3:50
Third				
Fourth		DLI	11/14 15:45	

I guarantee that as the client, or on behalf of client named, I have the authority to contract the above requested services. Should it be found that I do not have such authority, I agree to be personally liable for all costs and, if there should be action against me for this breach, reasonable attorneys' fees. It is understood that payment is expected to be cash with samples unless terms have been previously arranged. Terms are net 30 days; overdue accounts will be charged a liquidated damage fee of 2% per month (annually 24%) or $5.00 per month whichever is greater.
If payment is not made when due and a legitimate dispute exists concerning the product or services of Dellavalle Laboratory, Inc., it will be submitted to mediation under the Rules and Procedures of Creative Alternative to Litigation, Inc. (*cal*). If the dispute is not resolved in mediation, then the dispute will be submitted to binding arbitration through ***cal*** under its Rules and Procedures. The parties will equally bear the costs of mediation/arbitration. If, however, the mediator declares that no legitimate dispute exists, then debtor will pay all mediation and arbitration costs, and in the event of arbitration,reasonable attorneys' fees of Dellavalle Laboratory.

Invoicing Information:
Sampling hrs
Miles
Consulting

Shipping
$ In
$ Out

Amt Paid	Rec By	Check #	Date

Signature

Samples recieved in cooler with ice (coolant)
[ ] Yes [ ] No

11/14/23 15:45 23K0683

Shipping Information: Fed Ex ☐ UPS ☐ GSO ☐ Picked-Up ☒ Walk In ☐ DLI Sampler ☐
Ontrac ☐ PMS ☐ USPS ☐ Other ☐ _____

Shipping Container: Ice Chest ☒ None ☐ Box ☐

Refrigerant: Wet Ice: ☒ Blue Ice: ☐ None: ☐

Container(s) Received	1	2	3	4	5	6	7	8	9	10
Plastics:										
100 mL sterile plastic $Na_2S_2O_3$ (Green)										
250 mL unpreserved (White)										
250 mL HNO_3 (Red)										
250 mL H_2SO_4 (Yellow)										
500 mL unpreserved (White)										
500 mL HNO_3 (Red)										
500 mL H_2SO_4 (Yellow)										
1 L unpreserved (White)	1	1								
1 L unpreserved (BOD) (Purple)										
1 L HNO_3 (Red)										
1 L H_2SO_4 (Yellow)										
VOA Vials in sets of 3:										
40 mL VOA, HCl (Blue)										
40 mL VOA, $Na_2S_2O_3$ (Green)										
40 mL VOA, unpreserved (White)										
Glass:										
250 mL AG unpreserved (White)										
250 mL AG H_2SO_4 (Yellow)										
250 mL AG $Na_2S_2O_3$ (Green)										
250 mL AG $Na_2S_2O_3$ + MCAA										
500 mL glass unpreserved (White)										
500 mL AG HCl (Blue)										
1 L AG unpreserved (White)										
1 L AG H_2SO_4 (Yellow)										
1 L AG $Na_2S_2O_3$ (Green)										
1 L AG HCl (Blue)										
Special:										
Cr^{6+} w/Borate/HCO3/CO3										
Cyanide - 500 mL NaOH										
Asbestos - 2 - 1L P wrapped in foil										
500 mL or 1 L **AP** $Na_2S_2O_3$ (Green)										
1 L **AG** or **P** NaOH + ZnAc (Sulfide)										
Chlorite/Bromate 250 mL AG with EDA										
TTHM										
HAA5										
40 mL VOA, $Na_2S_2O_3$ + MCAA										